

Preparation of a New Spirobi[thieno[2,3-*c*]pyran] and Its Selective Mono- and Dimetalation: Application for the Preparation of Soluble Conjugated Oligothiophenes and Pyrene Derivatives

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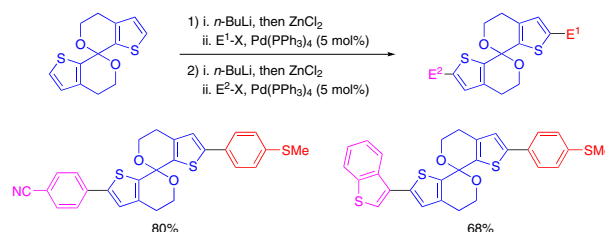
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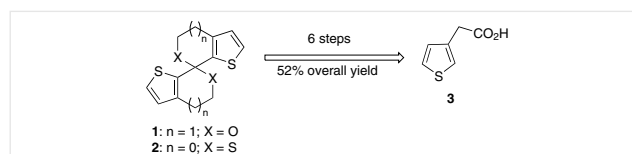
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Abstract 3-Thienylacetic acid was converted in six steps into a new spirobi[thieno[2,3-*c*]pyran] (overall yield: 52%). The spirobi[thieno[2,3-*c*]pyran] was selectively mono- or dimetalated with butyllithium and then transmetalated with zinc chloride; cross-coupling reaction with various aryl or heteroaryl bromides, including bromo-oligothiophenes, acid chlorides, and 1-bromopyrene, produced the corresponding spiro derivatives in high yields.

Key words lithiation, Negishi cross coupling, oligothiophene, organozinc reagent, spiro compound



Scheme 1 Retrosynthetic analysis for spiro compound **1**

The preparation of molecules bearing a quaternary center common to two ring systems (spiro compounds) has attracted much attention, since spiro compounds¹ are important products in various fields of organic chemistry such as drug discovery,² medicinal chemistry,³ asymmetric synthesis,⁴ natural product synthesis,⁵ and material science.⁶

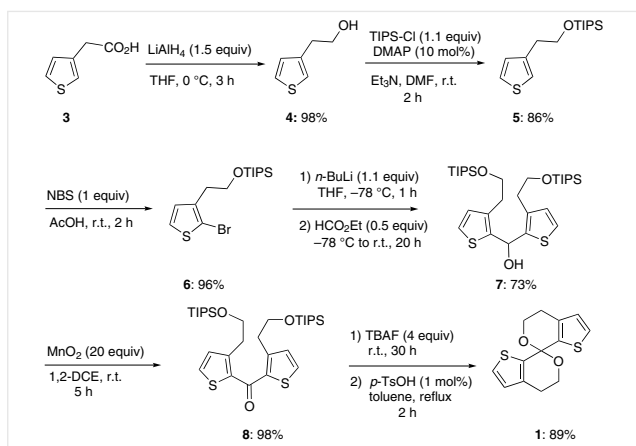
Due to our interest in the synthesis of soluble oligothiophenes,⁷ we envisioned the preparation of new spiro scaffolds containing two thiophene units such as **1** or **2**. Whereas preliminary experiments showed that the spiro compound **2** ($n = 0$, $X = S$) is not stable, the synthesis of **1** ($n = 1$, $X = O$) could be readily performed. Herein, we wish to report a straightforward synthesis of the new spirobi[thieno[2,3-*c*]pyran] **1** in six steps starting from 3-thienylacetic acid (**3**) in an overall yield of 52%, as well as its use as a conjunctive reagent⁸ to link thiophene or pyrenyl units together (Scheme 1).

Thus, the reduction of commercially available 3-thienylacetic acid (**3**) with lithium aluminum hydride,⁹ afforded quantitatively the corresponding alcohol **4**, which was silylated with triisopropylsilyl chloride¹⁰ leading to thiophene **5** in 86% yield. Regioselective bromination of **5** using

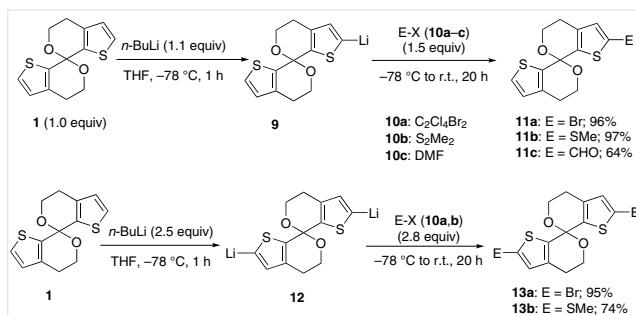
N-bromosuccinimide produced the 2,3-disubstituted thiophene **6** in 96% yield; Br/Li-exchange¹¹ on the thiophene **6** using butyllithium followed by the addition of ethyl formate afforded the symmetrical alcohol **7** in 73% yield. Oxidation¹² of this benzyl alcohol **7** with manganese dioxide gave ketone **8** in 98% yield. Desilylation of compound **8** with tetrabutylammonium fluoride¹³ in tetrahydrofuran led, after workup, to an intermediate diol that was refluxed in toluene in the presence of 1 mol% 4-toluenesulfonic acid to give the desired new and stable spiro compound **1** in 89% yield and 52% overall yield (Scheme 2).

We first examined the selective metalation and functionalization of the spiro molecule **1**. The lithiation of **1** with butyllithium (1.1 equiv) at -78°C in tetrahydrofuran led to selective monolithiation. The resulting thienyllithium **9** reacted directly with 1,2-dibromo-1,1,2,2-tetrachloroethane, dimethyl disulfide, or *N,N*-dimethylformamide furnishing the expected products **11a–c** in 64–97% yield (Scheme 3). Alternatively, the treatment of spiro compound **1** with butyllithium (2.5 equiv) at -78°C in tetrahydrofuran for one hour allowed dilithiation leading to the dilithiospiro compound **12**, which after bromination or methylthiolation furnished the desired difunctionalized spiro molecules **13a,b** in 74–95% yield.

Moreover, selective monolithiation of **1** with butyllithium followed by transmetalation with zinc chloride¹⁴ afforded thienylzinc reagent **14** (Scheme 4). This organozinc re-



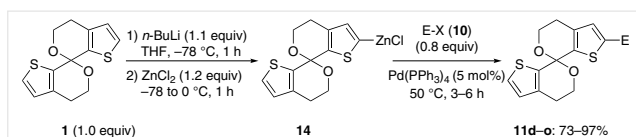
Scheme 2 Synthesis of new stable spirobi[thieno[2,3-c]pyran] **1**



Scheme 3 Mono- and dilithiation of spirobi[thieno[2,3-c]pyran] **1** followed by the addition of electrophiles **10a-c**

agent **14**¹⁵ underwent smooth cross coupling with various aryl bromides **10d-i** in the presence of 5 mol% tetrakis(triphenylphosphine)palladium(0) [Pd(PPh₃)₄] at 50 °C for 3–6 hours furnishing the arylated spiro products **11d-i** in 73–97% yield (Table 1, entries 1–6). Similarly, Negishi cross coupling¹⁶ with various heteroaryl bromides **10j-m** gave the corresponding monocoupling products **11j-m** in 85–96% yield (entries 7–10). Finally, palladium-catalyzed acylation¹⁷ reaction with the unsaturated acid chlorides **10n,o** under similar reaction conditions produced the corresponding spiro ketones **11n,o** in 89–94% yield (entries 11 and 12).

Notably, the introduction of two different electrophiles was also achieved. The metalation of **11b** (Scheme 3) with



Scheme 4 Monolithiation of spirobi[thieno[2,3-c]pyran] **1** followed by a transmetalation with zinc chloride and cross-coupling reaction with electrophiles **10d-o**

butyllithium (1.1 equiv) at –78 °C in tetrahydrofuran followed by transmetalation with zinc chloride and palladium-catalyzed cross-coupling with 4-bromobenzonitrile (**10g**) gave the expected product **15a** in 75% yield. Under similar reaction conditions, the metalation of **11i** (Table 1) followed by cross-coupling reaction with 4-bromobenzonitrile (**10g**) or 3-bromobenzothiophene (**10k**) afforded the

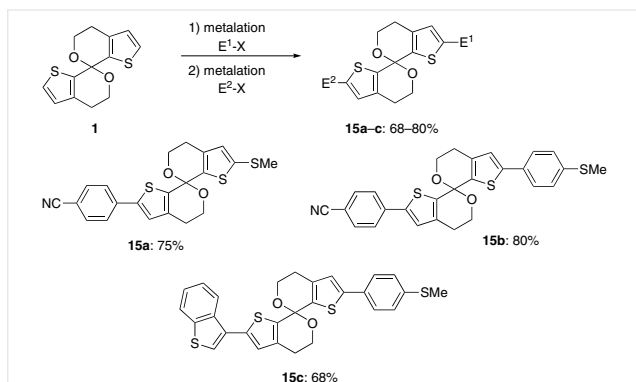
Table 1 Palladium-Catalyzed Negishi Cross-Coupling Reaction of Organozinc Reagent **14** with Various Electrophiles **10d-o**

Entry	Electrophile (E-X)	Product: yield ^{a,b}
1	10d : FG = CO ₂ Et	11d : 77%
2	10e : FG = COCH ₃	11e : 92%
3	10f : FG = CHO	11f : 96%
4	10g : FG = CN	11g : 97%
5	10h : FG = NO ₂	11h : 87%
6	10i : FG = SCH ₃	11i : 73%
7	 10j	 11j : 96%
8	 10k	 11k : 87%
9	 10l	 11l : 85%
10	 10m	 11m : 86%
11	 10n	 11n : 94%
12	 10o	 11o : 89%

^a General reaction conditions: Pd(PPh₃)₄ (5 mol%), ArBr or ArCOCl (0.8 equiv), organozinc reagent (1.0 equiv), THF, 50 °C.

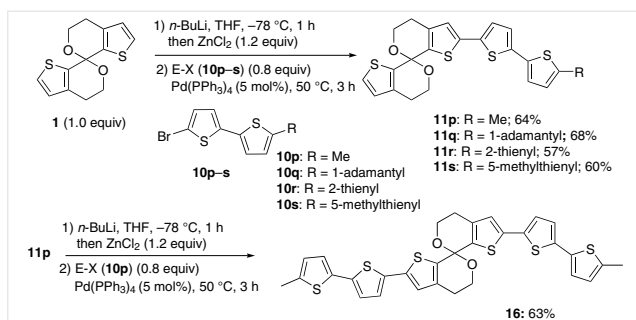
^b Yield after purification by flash column chromatography.

corresponding difunctionalized products **15b,c** in 68–80% yield (Scheme 5).



Scheme 5 Synthesis of unsymmetrically difunctionalized spiro compounds **15a–c**

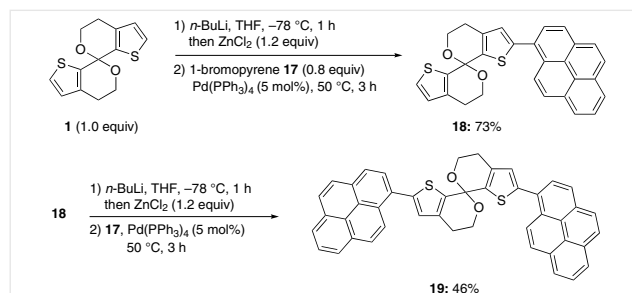
Remarkably, we anticipated that the cross coupling with oligothiophenes moieties should provide materials that are soluble in organic solvents. This could be confirmed, and selective mono-cross-coupling reaction of **1** with various bromothiophenes **10p–s** [5 mol% Pd(PPh₃)₄, THF, 50 °C, 3 h] gave the desired oligothiophenes **11p–s** in 57–68% yield. Interestingly, further treatment of **11p** with butyllithium and subsequent transmetalation with zinc chloride furnished, after palladium-catalyzed cross coupling with bromothiophene **10p**, the symmetrical disubstituted spiro molecule **16** in 63% yield (Scheme 6). The solubility of products **11p–s** and **16** is excellent in tetrahydrofuran and chloroform.



Scheme 6 Synthesis of soluble oligothiophenes **11p–s** and **16**

Furthermore, we have attached 1-pyrenyl moieties to the conjunctive reagent **1**. The pyrenyl group has found many applications in material science as organic electronics¹⁹ and the planar pyrenyl group often leads to products of low solubility in organic solvents. Thus, metalation of **1** (1. *n*-BuLi; 2. ZnCl₂) followed by standard Negishi cross coupling with 1-bromopyrene (**17**) provided the well-soluble mono-pyrenyl coupling product **18** in 73% yield (Scheme 7). Successfully, further metalation of **18** and subsequent pal-

ladium-catalyzed cross coupling with **17** gave the expected bis-pyrenyl product **19** in 46% yield. The solubility of **19** in organic solvents such as chloroform (3.7 g/L) and tetrahydrofuran (2.6 g/L) is acceptable.



Scheme 7 Synthesis of symmetrical pyrenyl-spiro molecule **19**

In summary, we have prepared a new conjunctive spiro molecule **1** in six steps from commercially available 3-thienylacetic acid with an overall yield of 52%. We have demonstrated the selective metalation of **1** and trapping with one or two electrophiles. Palladium-catalyzed cross coupling of organozinc reagents with oligothiophene units or 1-pyrenyl units provided soluble building blocks such as **16** and **19** that are of interest to materials science.

All reactions were carried out under an argon atmosphere in flame-dried glassware. Syringes, which were used to transfer anhydrous solvents or reagents, were purged with argon prior to use. THF was continuously refluxed and freshly distilled from Na benzophenone ketyl under N₂. Yields refer to isolated yields of compounds estimated to be >95% pure as determined by ¹H NMR (25 °C) and capillary GC. NMR spectra were recorded in a solution of CDCl₃ (residual CHCl₃: δ = 7.25 for ¹H NMR and δ = 77.0 for ¹³C NMR). Column chromatographic purification was performed using silica gel (0.040–0.063 mm, 230–400 mesh ASTM) from Merck. All reagents were obtained from commercial sources.

1 M Zinc Chloride in Tetrahydrofuran Solution

A dry and argon-flushed 250-mL Schlenk flask equipped with a magnetic stirring bar and a septum was charged with ZnCl₂ (13.6 g, 100 mmol). The salt was heated to 140 °C under high vacuum for 10 h. After cooling to 25 °C, anhyd THF (100 mL) was added and stirring was continued until the salt was dissolved completely (4 h).

2-(Thiophen-3-yl)ethan-1-ol (**4**)⁹

In a flame-dried Schlenk flask, equipped with a magnetic stirring bar, an argon inlet, and a septum, LiAlH₄ (5.7 g, 150 mmol, 1.5 equiv) was dissolved in THF (100 mL) at 0 °C. A solution of 3-thienylacetic acid (**3**, 14.2 g, 100 mmol, 1.0 equiv) in THF (50 mL) at 0 °C was added dropwise. The mixture was stirred for 3 h at 0 °C and then it was carefully quenched with EtOAc and EtOH. Subsequently, 1–2% NaOH (20 mL) was added and the mixture was stirred for 1 h, allowing aluminum hydroxide to precipitate as a granular solid. Finally, the mixture was filtered over Celite. The collected organic layers were dried (MgSO₄), and the solvent was evaporated in vacuo to give the pure alcohol **4** (12.5 g, 98%) as a pale yellow oil.

Triisopropyl[2-(thiophen-3-yl)ethoxy]silane (5)

A dry, argon-flushed Schlenk flask equipped with a magnetic stirring bar and a septum was charged with 2-(thiophen-3-yl)ethan-1-ol (**4**, 12.3 g, 96 mmol, 1.0 equiv) and DMF (100 mL) at r.t. Subsequently, Et₃N (10.7 g, 105 mmol, 1.1 equiv), DMAP (1.17 g, 9.6 mmol), and TIPSCl (20.3 g, 105 mmol, 1.1 equiv) were added. The resulting mixture was stirred for 2 h at r.t. and then it was quenched with water and extracted with Et₂O (5 × 60 mL). The combined organic layers were dried (MgSO₄) and the solvent was evaporated in vacuo. The crude residue was purified by flash column chromatography (silica gel pre-neutralized with Et₃N, isohexane–EtOAc, 99:1) and gave the pure silylated compound **5** (23.4 g, 86%) as a pale yellow oil.

IR (ATR): 2941, 2864, 1462, 1100, 881, 769, 677 cm⁻¹.

¹H NMR (300 MHz, CDCl₃): δ = 7.18–7.27 (m, 1 H), 6.93–7.04 (m, 2 H), 3.88 (t, *J* = 6.9 Hz, 2 H), 2.88 (t, *J* = 6.9 Hz, 2 H), 0.77–1.35 (m, 3 H), 1.04 (d, *J* = 4.2 Hz, 18 H).

¹³C NMR (75 MHz, CDCl₃): δ = 139.5, 128.7, 124.9, 121.1, 64.0, 34.0, 18.0, 12.0.

MS (EI): *m/z* (%) = 241 ([M – C₃H₇]⁺, 100), 213 (15), 171 (13), 111 (28), 75 (9), 43 (31).

HRMS (EI): *m/z* [M – C₃H₇] calcd for C₁₂H₂₁OSSi: 241.1082; found: 241.1061.

[2-(2-Bromothiophen-3-yl)ethoxy]triisopropylsilane (6)

To a solution of thiophene **5** (15.6 g, 54.8 mmol, 1.0 equiv) in AcOH (27 mL), NBS (9.76 g, 54.8 mmol, 1.0 equiv) was added at 0 °C and the mixture was stirred at r.t. for 2 h. The mixture was quenched with water (100 mL) and then it was neutralized with sat. aq NaHCO₃ solution. The resulting solution was extracted with Et₂O (4 × 50 mL). The combined organic phases were dried (Na₂SO₄) and concentrated in vacuo. The crude residue obtained was purified by flash column chromatography (silica gel pre-neutralized with Et₃N, isohexane–EtOAc, 99:1) to give bromo compound **6** (19.1 g, 96%) as a pale yellow oil.

IR (ATR): 2941, 2864, 1462, 1103, 881, 679 cm⁻¹.

¹H NMR (300 MHz, CDCl₃): δ = 7.17 (d, *J* = 5.5 Hz, 1 H), 6.88 (d, *J* = 5.5 Hz, 1 H), 3.85 (t, *J* = 6.9 Hz, 2 H), 2.84 (t, *J* = 6.9 Hz, 2 H), 0.73–1.35 (m, 3 H), 1.04 (d, *J* = 4.2 Hz, 18 H).

¹³C NMR (75 MHz, CDCl₃): δ = 138.7, 129.1, 125.0, 109.8, 62.7, 33.1, 17.9, 11.9.

MS (EI): *m/z* (%) = 321 ([M – C₃H₇]⁺, 100), 319 ([M – C₃H₇]⁺, 85), 290 (19), 190 (15), 169 (24), 132 (11), 110 (22), 43 (10).

HRMS (EI): *m/z* [M – C₃H₇] calcd for C₁₂H₂₀BrOSSi: 319.0188; found: 319.0167.

Bis[3-[2-(triisopropylsiloxy)ethyl]thiophen-2-yl]methanol (7)

A dry, argon-flushed Schlenk flask equipped with a magnetic stirring bar and a septum was charged with bromothiophene **6** (3.0 g, 8.25 mmol, 1.0 equiv) dissolved in THF (40 mL). 2.48 M *n*-BuLi in hexane (3.66 mL, 9.08 mmol, 1.1 equiv) was added dropwise to this solution at –78 °C and the resulting mixture was stirred for 1 h; ethyl formate (0.30 g, 4.12 mmol, 0.5 equiv) was added and the mixture was allowed to warm up to r.t. and stirred overnight. The resulting mixture was quenched with sat. aq NH₄Cl (30 mL) and extracted with Et₂O (3 × 50 mL). The combined organic phases were dried (Na₂SO₄) and concentrated in vacuo. The crude residue obtained was purified by flash

column chromatography (silica gel pre-neutralized with Et₃N, isohexane–EtOAc, 90:10) to give the pure alcohol product **7** (1.8 g, 73%) as a pale yellow solid; mp 40.0–41.3 °C.

IR (ATR): 3432, 2940, 2865, 1462, 1084, 1032, 882, 740, 684 cm⁻¹.

¹H NMR (300 MHz, CDCl₃): δ = 7.15 (d, *J* = 5.0 Hz, 2 H), 6.87 (d, *J* = 5.0 Hz, 2 H), 6.34 (s, 1 H), 4.41 (s, 1 H), 3.64–3.97 (m, 4 H), 2.70–3.04 (m, 4 H), 0.61–1.46 (m, 6 H), 1.00 (d, *J* = 4.7 Hz, 36 H).

¹³C NMR (75 MHz, CDCl₃): δ = 143.0, 135.4, 128.9, 123.8, 64.5, 63.6, 32.0, 17.8, 11.9.

MS (ESI): *m/z* (%) = 579 ([M – OH]⁺, 100), 295 (4).

HRMS (ESI): *m/z* [M – OH] calcd for C₃₁H₅₅O₂S₂Si₂: 579.3182; found: 579.3173.

Bis[3-[2-(triisopropylsiloxy)ethyl]thiophen-2-yl]methanone (8)

Activated MnO₂ (34.1 g, 392.3 mmol, 20 equiv) was added to a solution of alcohol **7** (11.7 g, 19.59 mmol, 1.0 equiv) in anhyd 1,2-dichloroethane (400 mL); the mixture was stirred at r.t. for 5 h. The resulting solution was passed through Celite pad and washed with CHCl₃ (5 × 50 mL). The removal of solvent gave the analytically pure product **8** (11.4 g, 98%) as a pale yellow oil.

IR (ATR): 2941, 2864, 1619, 1408, 1096, 880, 677 cm⁻¹.

¹H NMR (400 MHz, CDCl₃): δ = 7.43 (d, *J* = 4.8 Hz, 2 H), 7.14 (d, *J* = 4.8 Hz, 2 H), 3.96 (t, *J* = 6.4 Hz, 4 H), 3.16 (t, *J* = 6.4 Hz, 4 H), 0.81–1.21 (m, 6 H), 1.04 (d, *J* = 5.4 Hz, 36 H).

¹³C NMR (100 MHz, CDCl₃): δ = 181.8, 147.2, 136.4, 132.0, 129.3, 63.6, 33.7, 18.0, 12.0.

MS (EI): *m/z* (%) = 576 ([M – H₂O]⁺, 6), 551 (100), 365 (85), 131 (60), 103 (86), 74 (65), 43 (87).

HRMS (EI): *m/z* [M – H₂O] calcd for C₃₁H₅₂O₂S₂Si₂: 576.2947; found: 576.2954.

4,4',5,5'-Tetrahydro-7,7'-spirobi[thieno[2,3-c]pyran (1)

A dry, argon-flushed Schlenk flask equipped with a magnetic stirring bar and a septum was charged with ketone **8** (5.0 g, 8.4 mmol, 1.0 equiv) in anhyd THF (80 mL) and cooled to 0 °C. Then 1 M TBAF in THF (30.54 mL, 33.6 mmol, 4.0 equiv) was added and the mixture was stirred at 0 to 25 °C for 30 h. Removal of the solvent afforded the crude mixture which was quenched with sat. aq NH₄Cl solution (20 mL) and extracted with EtOAc (4 × 30 mL). The combined organic phases were dried (Na₂SO₄) and concentrated in vacuo. The obtained crude diol was treated with *p*-TsOH (15.9 mg, 0.084 mmol, 0.01 equiv) in toluene (200 mL) and refluxed for 2 h; the mixture was cooled to r.t. After removal of the solvent, the crude residue obtained was purified by flash column chromatography (silica gel pre-neutralized with Et₃N, isohexane–EtOAc, 85:15) to give the pure spiro product **1** (1.97 g, 89%) as a pale yellow solid; mp 156.7–157.6 °C.

IR (ATR): 2934, 2876, 1554, 1277, 1244, 1158, 1041, 963, 932, 730, 715 cm⁻¹.

¹H NMR (600 MHz, CDCl₃): δ = 7.24 (d, *J* = 5.2 Hz, 2 H), 6.80 (d, *J* = 5.2 Hz, 2 H), 4.27 (td, *J* = 11.8, 3.0 Hz, 2 H), 4.06 (dd, *J* = 11.3, 5.7 Hz, 2 H), 2.95–3.06 (m, 2 H), 2.63 (dd, *J* = 16.2, 3.0 Hz, 2 H).

¹³C NMR (150 MHz, CDCl₃): δ = 137.5, 136.3, 125.8, 125.7, 94.7, 60.0, 25.8.

MS (EI): *m/z* (%) = 264 (M⁺, 10), 235 (17), 234 (100), 201 (23), 109 (6), 97 (6).

HRMS (EI): *m/z* [M]⁺ calcd for C₁₃H₁₂O₂S₂: 264.0279; found: 264.0268.

Reaction of Organolithium Reagent **9** with Various Electrophiles **10a–c**; Typical Procedure **1** (TP1)

A dry, argon-flushed Schlenk flask equipped with a magnetic stirring bar and a septum was charged with spiro compound **1** (0.38 mmol, 1.0 equiv) in anhyd THF (3 mL) and cooled to -78°C for 10 min. 2.48 M *n*-BuLi in hexane (0.42 mmol, 1.1 equiv) was added and the mixture was stirred at -78°C for 1 h until GC analysis of reaction aliquots showed full consumption of the starting material. Then the electrophile **10a–c** (0.6 mmol, 1.5 equiv) was added and the mixture was allowed to warm up to r.t. and stirred overnight. The mixture was quenched with sat. aq NH_4Cl solution (5 mL) and extracted with EtOAc (4×15 mL). The combined organic phases were dried (Na_2SO_4) and concentrated in vacuo. The crude residue obtained was purified by flash column chromatography (silica gel pre-neutralized with Et_3N , isohexane–EtOAc) to give the analytically pure spiro product **11a–c**.

2-Bromo-4,4',5,5'-tetrahydro-7,7'-spirobi[thieno[2,3-*c*]pyran] (**11a**)

The reaction of spiro compound **1** (100 mg, 0.38 mmol, 1.0 equiv) in anhyd THF (3 mL) with 2.48 M *n*-BuLi in hexane (0.17 mL, 0.42 mmol, 1.1 equiv) and 1,2-dibromo-1,1,2,2-tetrachloroethane (**10a**, 185 mg, 0.57 mmol, 1.5 equiv) was performed according to TP1 overnight. Flash column chromatographic purification (silica gel pre-neutralized with Et_3N , isohexane–EtOAc, 85:15) furnished **11a** (125 mg, 96%) as a pale yellow solid; mp 153.9 – 155.0°C .

IR (ATR): 2924, 2885, 1441, 1242, 1035, 974, 929, 816, 714 cm^{-1} .

^1H NMR (400 MHz, CDCl_3): δ = 7.19 (d, J = 4.7 Hz, 1 H), 6.73 (d, J = 4.7 Hz, 1 H), 6.70 (s, 1 H), 4.09–4.26 (m, 2 H), 3.96 (td, J = 12.4, 5.7 Hz, 2 H), 2.80–2.97 (m, 2 H), 2.40–2.60 (m, 2 H).

^{13}C NMR (100 MHz, CDCl_3): δ = 139.3, 137.1, 136.8, 136.6, 128.8, 126.0, 126.0, 114.0, 94.1, 60.1, 59.9, 25.9, 25.4.

MS (EI): m/z (%) = 344 (M^+ , 12), 342 (M^+ , 14), 314 (100), 312 (97), 234 (20), 109 (25), 69 (28), 43 (38).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{13}\text{H}_{11}\text{BrO}_2\text{S}_2$: 341.9384; found: 341.9386.

2-(Methylthio)-4,4',5,5'-tetrahydro-7,7'-spirobi[thieno[2,3-*c*]pyran] (**11b**)

The reaction of spiro compound **1** (100 mg, 0.38 mmol, 1.0 equiv) in anhyd THF (3 mL) with 2.48 M *n*-BuLi in hexane (0.17 mL, 0.42 mmol, 1.1 equiv) and dimethyl disulfide (**10b**, 53 mg, 0.57 mmol, 1.5 equiv) was performed according to TP1 overnight. Flash column chromatographic purification (silica gel pre-neutralized with Et_3N , isohexane–EtOAc, 80:20) furnished **11b** (114 mg, 97%) as a pale yellow solid; mp 103.3 – 105.3°C .

IR (ATR): 2967, 2914, 1556, 1424, 1273, 1151, 1034, 976, 932, 718 cm^{-1} .

^1H NMR (400 MHz, CDCl_3): δ = 7.18 (d, J = 5.0 Hz, 1 H), 6.73 (d, J = 5.0 Hz, 1 H), 6.68 (s, 1 H), 4.12–4.25 (m, 2 H), 3.97 (td, J = 10.5, 6.1 Hz, 2 H), 2.79–2.97 (m, 2 H), 2.42–2.60 (m, 2 H), 2.36 (s, 3 H).

^{13}C NMR (100 MHz, CDCl_3): δ = 139.3, 138.6, 137.1, 136.6, 136.5, 129.0, 125.9, 125.8, 94.4, 60.1, 60.0, 25.9, 25.6, 21.6.

MS (EI): m/z (%) = 310 (M^+ , 27), 282 (18), 281 (27), 280 (100), 265 (19), 171 (10), 97 (14).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{14}\text{H}_{14}\text{O}_2\text{S}_3$: 310.0156; found: 310.0165.

4,4',5,5'-Tetrahydro-7,7'-spirobi[thieno[2,3-*c*]pyran]-2-carbaldehyde (**11c**)

The reaction of spiro compound **1** (100 mg, 0.38 mmol, 1.0 equiv) in anhyd THF (3 mL) with 2.48 M *n*-BuLi in hexane (0.17 mL, 0.42 mmol, 1.1 equiv) and DMF **10c** (42 mg, 0.57 mmol, 1.5 equiv) was performed according to TP1 overnight. Flash column chromatographic purification (silica gel pre-neutralized with Et_3N , isohexane–EtOAc, 50:50) furnished **11c** (71 mg, 64%) as a pale yellow solid; mp 171.6 – 172.8°C .

IR (ATR): 2925, 2852, 1666, 1272, 1233, 1137, 1023, 969, 937, 714 cm^{-1} .

^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ = 9.90 (s, 1 H), 7.81 (s, 1 H), 7.53 (d, J = 5.1 Hz, 1 H), 6.92 (d, J = 5.1 Hz, 1 H), 3.91–4.22 (m, 4 H), 2.62–2.96 (m, 4 H).

^{13}C NMR (100 MHz, $\text{DMSO}-d_6$): δ = 185.1, 147.1, 142.8, 138.3, 137.3, 136.9, 136.4, 127.0, 126.6, 94.4, 60.0, 59.7, 25.7, 25.5.

MS (EI): m/z (%) = 292 (M^+ , 6), 262 (100), 229 (17), 190 (7), 109 (5), 65 (4).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{14}\text{H}_{12}\text{O}_3\text{S}_2$: 292.0228; found: 292.0219.

2,2'-Dibromo-4,4',5,5'-tetrahydro-7,7'-spirobi[thieno[2,3-*c*]pyran] (**13a**)

A dry, argon-flushed Schlenk flask equipped with a magnetic stirring bar and a septum was charged with spiro compound **1** (200 mg, 0.76 mmol, 1.0 equiv) in anhyd THF (6 mL) and cooled to -78°C for 10 min. 2.48 M *n*-BuLi in hexane (0.76 mL, 1.89 mmol, 2.5 equiv) was added and the mixture was stirred at -78°C for 1 h until GC analysis of reaction aliquots showed full consumption of the starting material. Then 1,2-dibromo-1,1,2,2-tetrachloroethane (**10a**, 691 mg, 2.12 mmol, 2.8 equiv) was added and the mixture was allowed to warm up to r.t. and stirred overnight. The mixture was quenched with sat. aq NH_4Cl solution (5 mL) and extracted with EtOAc (4×15 mL). The combined organic phases were dried (Na_2SO_4) and concentrated in vacuo. The crude residue obtained was purified by flash column chromatography (silica gel pre-neutralized with Et_3N , isohexane–EtOAc, 90:10) to furnish **13a** (302 mg, 95%) as a pale yellow solid; mp 197.0 – 197.4°C .

IR (ATR): 2991, 2952, 1558, 1440, 1240, 1146, 1033, 977, 928, 815 cm^{-1} .

^1H NMR (400 MHz, CDCl_3): δ = 6.77 (s, 2 H), 4.22 (td, J = 11.7, 3.0 Hz, 2 H), 4.01 (dd, J = 11.2, 5.6 Hz, 2 H), 2.82–2.99 (m, 2 H), 2.53 (dd, J = 16.4, 3.0 Hz, 2 H).

^{13}C NMR (100 MHz, CDCl_3): δ = 138.5, 137.1, 128.8, 114.3, 93.5, 60.0, 25.4.

MS (EI): m/z (%) = 422 (M^+ , 16), 420 (M^+ , 8), 394 (55), 392 (100), 390 (45), 313 (63), 311 (55), 232 (12), 109 (9).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{13}\text{H}_{10}\text{Br}_2\text{O}_2\text{S}_2$: 419.8489; found: 419.8485.

2,2'-Bis(methylthio)-4,4',5,5'-tetrahydro-7,7'-spirobi[thieno[2,3-*c*]pyran] (**13b**)

A dry, argon-flushed Schlenk flask equipped with a magnetic stirring bar and a septum was charged with spiro compound **1** (100 mg, 0.38 mmol, 1.0 equiv) in anhyd THF (4 mL) and cooled to -78°C for 10 min. 2.48 M *n*-BuLi in hexane (0.38 mL, 0.95 mmol, 2.5 equiv) was added and the mixture was stirred at -78°C for 1 h until GC analysis of reaction aliquots showed full consumption of the starting material. Then dimethyl disulfide (**10b**, 100 mg, 1.1 mmol, 2.8 equiv) was added and the mixture was allowed to warm up to r.t. and stirred overnight. The mixture was quenched with sat. aq NH_4Cl solution (5 mL) and extracted with EtOAc (4×15 mL). The combined organic phases were dried (Na_2SO_4) and concentrated in vacuo. The crude residue ob-

tained was purified by flash column chromatography (silica gel pre-neutralized with Et₃N, isohexane–EtOAc, 80:20) to furnish **13b** as a white solid (100 mg, 74%); mp 114.6–116.7 °C.

IR (ATR): 2965, 2920, 2854, 1562, 1446, 1271, 1031, 930, 794 cm⁻¹.

¹H NMR (400 MHz, CDCl₃): δ = 6.76 (s, 2 H), 4.25 (td, *J* = 11.7, 3.0 Hz, 2 H), 4.04 (dd, *J* = 11.3, 5.9 Hz, 2 H), 2.86–2.99 (m, 2 H), 2.54 (dd, *J* = 16.1, 3.0 Hz, 2 H), 2.46 (s, 6 H).

¹³C NMR (100 MHz, CDCl₃): δ = 138.8, 138.7, 136.7, 129.0, 94.0, 60.0, 25.6, 21.6.

MS (EI): *m/z* (%) = 356 (M⁺, 32), 326 (100), 311 (15), 279 (70), 97 (9), 69 (8).

HRMS (EI): *m/z* [M]⁺ calcd for C₁₅H₁₆O₄S₄: 356.0033; found: 356.0029.

Palladium-Catalyzed Cross-Coupling Reactions of Organozinc Reagent **14** with Various Electrophiles **10d–o**; Typical Procedure 2 (TP2)

A dry, argon-flushed Schlenk flask equipped with a magnetic stirring bar and a septum was charged with spiro compound **1** (0.38 mmol, 1.0 equiv) in anhyd THF (3 mL) and cooled to –78 °C for 10 min. 2.48 M *n*-BuLi in hexane (0.42 mmol, 1.1 equiv) was added and the mixture was stirred at –78 °C for 1 h until GC analysis of reaction aliquots showed full consumption of the starting material. 1 M ZnCl₂ in THF solution (0.46 mmol, 1.2 equiv) was added and the mixture was stirred at –78 to 0 °C for 1 h. An aryl bromide or acid chloride **10d–o** (0.30 mmol, 0.8 equiv) and Pd(PPh₃)₄ (0.015 mmol, 0.05 equiv, 5 mol%) were added and the mixture was stirred at 50 °C for 3–6 h. The mixture was quenched with sat. aq NH₄Cl solution (5 mL) and extracted with EtOAc (4 × 15 mL). The combined organic phases were dried (Na₂SO₄) and concentrated in vacuo. The crude residue obtained was purified by flash column chromatography (silica gel pre-neutralized with Et₃N, isohexane–EtOAc) to give the analytically pure spiro product **11d–o**.

Ethyl 4-(4,4',5,5'-Tetrahydro-7,7'-spirobi[thieno[2,3-*c*]pyran]-2-yl)benzoate (**11d**)

Cross coupling of spiro compound **1** (100 mg, 0.38 mmol, 1.0 equiv) was performed according to TP2. The intermediate zinc reagent was prepared using 2.48 M *n*-BuLi in hexane (0.17 mL, 0.42 mmol, 1.1 equiv) and 1 M ZnCl₂ in THF solution (0.46 mL, 0.46 mmol, 1.2 equiv) and cross coupling was performed using the aryl bromide **10d** (69 mg, 0.30 mmol, 0.8 equiv) and Pd(PPh₃)₄ (17 mg, 0.015 mmol, 0.05 equiv) at 50 °C for 3 h. Flash column chromatographic purification (silica gel pre-neutralized with Et₃N, isohexane–EtOAc, 70:30) furnished **11d** (95 mg, 77%) as a pale yellow solid; mp 183.6–184.7 °C.

IR (ATR): 3111, 2981, 2922, 2884, 1712, 1606, 1273, 1027, 971, 766, 696 cm⁻¹.

¹H NMR (400 MHz, CDCl₃): δ = 7.92 (d, *J* = 8.2 Hz, 2 H), 7.50 (d, *J* = 8.4 Hz, 2 H), 7.19 (d, *J* = 5.1 Hz, 1 H), 7.05 (s, 1 H), 6.76 (d, *J* = 5.1 Hz, 1 H), 4.14–4.34 (m, 4 H), 4.02 (dd, *J* = 11.0, 5.7 Hz, 2 H), 2.87–3.03 (m, 2 H), 2.57 (dt, *J* = 16.3, 3.8 Hz, 2 H), 1.31 (t, *J* = 7.1 Hz, 3 H).

¹³C NMR (100 MHz, CDCl₃): δ = 166.2, 143.2, 138.6, 138.3, 137.5, 137.0, 136.6, 130.2, 129.4, 126.0, 125.9, 125.4, 123.3, 94.6, 61.0, 60.1, 60.0, 25.9, 25.8, 14.4.

MS (EI): *m/z* (%) = 412 (M⁺, 11), 383 (23), 382 (100), 349 (22), 171 (6), 147 (6).

HRMS (EI): *m/z* [M]⁺ calcd for C₂₂H₂₀O₄S₂: 412.0803; found: 412.0798.

1-[4-(4,4',5,5'-Tetrahydro-7,7'-spirobi[thieno[2,3-*c*]pyran]-2-yl)phenyl]ethan-1-one (**11e**)

Cross coupling of spiro compound **1** (100 mg, 0.38 mmol, 1.0 equiv) was performed according to TP2. The intermediate zinc reagent was prepared using 2.48 M *n*-BuLi in hexane (0.17 mL, 0.42 mmol, 1.1 equiv) and 1 M ZnCl₂ in THF solution (0.46 mL, 0.46 mmol, 1.2 equiv) and cross coupling was performed using the aryl bromide **10e** (60 mg, 0.30 mmol, 0.8 equiv) and Pd(PPh₃)₄ (17 mg, 0.015 mmol, 0.05 equiv) at 50 °C for 4 h. Flash column chromatographic purification (silica gel pre-neutralized with Et₃N, isohexane–EtOAc, 50:50) furnished **11e** as a yellow solid (105 mg, 92%); mp 155.6–158.2 °C.

IR (ATR): 3104, 3010, 2909, 1682, 1602, 1272, 1158, 1034, 968, 824, 718 cm⁻¹.

¹H NMR (400 MHz, CDCl₃): δ = 7.94 (d, *J* = 8.4 Hz, 2 H), 7.62 (d, *J* = 8.4 Hz, 2 H), 7.30 (d, *J* = 5.1 Hz, 1 H), 7.17 (s, 1 H), 6.86 (d, *J* = 5.1 Hz, 1 H), 4.25–4.41 (m, 2 H), 4.12 (dd, *J* = 11.3, 5.8 Hz, 2 H), 2.98–3.12 (m, 2 H), 2.68 (dt, *J* = 16.2, 4.0 Hz, 2 H), 2.61 (s, 3 H).

¹³C NMR (100 MHz, CDCl₃): δ = 197.3, 143.0, 138.8, 138.6, 137.6, 137.0, 136.6, 135.9, 129.1, 126.0, 125.9, 125.6, 123.5, 94.6, 60.2, 60.0, 26.6, 25.9, 25.8.

MS (EI): *m/z* (%) = 382 (M⁺, 14), 354 (14), 353 (22), 352 (100), 319 (30), 154 (8), 43 (37).

HRMS (EI): *m/z* [M]⁺ calcd for C₂₁H₁₈O₃S₂: 382.0697; found: 382.0690.

4-(4,4',5,5'-Tetrahydro-7,7'-spirobi[thieno[2,3-*c*]pyran]-2-yl)benzaldehyde (**11f**)

Cross coupling of spiro compound **1** (100 mg, 0.38 mmol, 1.0 equiv) was performed according to TP2. The intermediate zinc reagent was prepared using 2.48 M *n*-BuLi in hexane (0.17 mL, 0.42 mmol, 1.1 equiv) and 1 M ZnCl₂ in THF solution (0.46 mL, 0.46 mmol, 1.2 equiv) and cross coupling was performed using the aryl bromide **10f** (56 mg, 0.30 mmol, 0.8 equiv) and Pd(PPh₃)₄ (17 mg, 0.015 mmol, 0.05 equiv) at 50 °C for 4 h. Flash column chromatographic purification (silica gel pre-neutralized with Et₃N, isohexane–EtOAc, 50:50) furnished **11f** as a yellow solid (106 mg, 96%); mp 129.0–130.8 °C.

IR (ATR): 3113, 2909, 2833, 2734, 1696, 1601, 1558, 1277, 1159, 1036, 968, 821, 717 cm⁻¹.

¹H NMR (400 MHz, DMSO-*d*₆): δ = 9.97 (s, 1 H), 7.89 (d, *J* = 8.2 Hz, 2 H), 7.81 (d, *J* = 8.2 Hz, 2 H), 7.49–7.54 (m, 2 H), 6.91 (d, *J* = 5.1 Hz, 1 H), 3.99–4.13 (m, 4 H), 2.80–2.94 (m, 2 H), 2.62–2.72 (m, 2 H).

¹³C NMR (100 MHz, DMSO-*d*₆): δ = 192.7, 141.9, 139.5, 139.3, 138.4, 137.2, 137.1, 135.6, 130.8, 126.8, 126.6, 126.1, 125.3, 94.4, 59.9, 59.8, 25.8, 25.7.

MS (EI): *m/z* (%) = 368 (M⁺, 14), 338 (100), 310 (9), 305 (27), 171 (6).

HRMS (EI): *m/z* [M]⁺ calcd for C₂₀H₁₆O₃S₂: 368.0541; found: 368.0538.

4-(4,4',5,5'-Tetrahydro-7,7'-spirobi[thieno[2,3-*c*]pyran]-2-yl)benzonitrile (**11g**)

Cross coupling of spiro compound **1** (100 mg, 0.38 mmol, 1.0 equiv) was performed according to TP2. The intermediate zinc reagent was prepared using 2.48 M *n*-BuLi in hexane (0.17 mL, 0.42 mmol, 1.1 equiv) and 1 M ZnCl₂ in THF solution (0.46 mL, 0.46 mmol, 1.2 equiv) and cross coupling was performed using the aryl bromide **10g** (55 mg, 0.30 mmol, 0.8 equiv) and Pd(PPh₃)₄ (17 mg, 0.015 mmol, 0.05 equiv) at 50 °C for 5 h. Flash column chromatographic purification (silica gel pre-neutralized with Et₃N, isohexane–EtOAc, 67:33) furnished **11g** (106 mg, 97%) as a pale yellow solid; mp 152.2–153.2 °C.

IR (ATR): 3112, 2939, 2906, 2832, 2226, 1604, 1276, 1158, 1034, 968, 930, 818, 718 cm^{-1} .

^1H NMR (400 MHz, CDCl_3): δ = 7.54–7.64 (m, 4 H), 7.28 (d, J = 5.1 Hz, 1 H), 7.13 (s, 1 H), 6.84 (d, J = 5.1 Hz, 1 H), 4.22–4.37 (m, 2 H), 4.10 (dd, J = 11.1, 5.7 Hz, 2 H), 2.95–3.10 (m, 2 H), 2.66 (ddd, J = 16.2, 5.7, 3.2 Hz, 2 H).

^{13}C NMR (100 MHz, CDCl_3): δ = 142.1, 139.5, 138.5, 137.8, 136.8, 136.7, 132.7, 126.0, 126.0, 126.0, 124.0, 118.8, 110.8, 94.5, 60.2, 60.0, 25.9, 25.8.

MS (EI): m/z (%) = 365 (M^+ , 11), 335 (100), 302 (24), 146 (5).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{20}\text{H}_{15}\text{NO}_2\text{S}_2$: 365.0544; found: 365.0533.

2-(4-Nitrophenyl)-4,4',5,5'-tetrahydro-7,7'-spirobi[thieno[2,3-c]pyran] (11h)

Cross coupling of spiro compound **1** (100 mg, 0.38 mmol, 1.0 equiv) was performed according to TP2. The intermediate zinc reagent was prepared using 2.48 M *n*-BuLi in hexane (0.17 mL, 0.42 mmol, 1.1 equiv) and 1 M ZnCl_2 in THF solution (0.46 mL, 0.46 mmol, 1.2 equiv) and cross coupling was performed using the aryl bromide **10h** (61 mg, 0.30 mmol, 0.8 equiv) and $\text{Pd}(\text{PPh}_3)_4$ (17 mg, 0.015 mmol, 0.05 equiv) at 50 °C for 4 h. Flash column chromatographic purification (silica gel pre-neutralized with Et_3N , isohexane–EtOAc, 65:35) furnished **11h** as a yellow solid (100 mg, 87%); mp 223.4–225.2 °C.

IR (ATR): 3109, 2970, 2876, 1593, 1506, 1335, 1274, 1160, 1108, 1036, 971, 932, 850, 750, 722, 687 cm^{-1} .

^1H NMR (400 MHz, CDCl_3): δ = 8.19 (d, J = 8.8 Hz, 2 H), 7.64 (d, J = 8.8 Hz, 2 H), 7.28 (d, J = 5.1 Hz, 1 H), 7.19 (s, 1 H), 6.85 (d, J = 5.1 Hz, 1 H), 4.30 (dddd, J = 14.8, 11.9, 11.3, 3.1 Hz, 2 H), 4.10 (dd, J = 11.2, 5.7 Hz, 2 H), 2.95–3.12 (m, 2 H), 2.61–2.74 (m, 2 H).

^{13}C NMR (100 MHz, CDCl_3): δ = 146.8, 141.6, 140.4, 140.2, 137.9, 136.7, 126.1, 126.0, 125.9, 124.6, 124.4, 94.5, 60.2, 60.0, 25.9, 25.8.

MS (EI): m/z (%) = 385 (M^+ , 11), 327 (6), 322 (22), 276 (7), 137 (5).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{19}\text{H}_{15}\text{NO}_4\text{S}_2$: 385.0442; found: 385.0436.

2-[4-(Methylthio)phenyl]-4,4',5,5'-tetrahydro-7,7'-spirobi[thieno[2,3-c]pyran] (11i)

Cross coupling of spiro compound **1** (100 mg, 0.38 mmol, 1.0 equiv) was performed according to TP2. The intermediate zinc reagent was prepared using 2.48 M *n*-BuLi in hexane (0.17 mL, 0.42 mmol, 1.1 equiv) and 1 M ZnCl_2 in THF solution (0.46 mL, 0.46 mmol, 1.2 equiv) and cross coupling was performed using the aryl bromide **10i** (61 mg, 0.30 mmol, 0.8 equiv) and $\text{Pd}(\text{PPh}_3)_4$ (17 mg, 0.015 mmol, 0.05 equiv) at 50 °C for 6 h. Flash column chromatographic purification (silica gel pre-neutralized with Et_3N , isohexane–EtOAc, 70:30) furnished **11i** (85 mg, 73%) as a pale yellow solid; mp 173.4–174.2 °C.

IR (ATR): 3101, 2982, 2955, 2881, 1559, 1502, 1462, 1278, 1162, 1037, 973, 926, 812, 729 cm^{-1} .

^1H NMR (400 MHz, CDCl_3): δ = 7.46 (d, J = 8.2 Hz, 2 H), 7.29 (d, J = 5.1 Hz, 1 H), 7.23 (d, J = 8.20 Hz, 2 H), 7.01 (s, 1 H), 6.85 (d, J = 5.1 Hz, 1 H), 4.33 (dddd, J = 15.2, 12.0, 11.8, 3.0 Hz, 2 H), 4.04–4.18 (m, 2 H), 2.97–3.13 (m, 2 H), 2.66 (td, J = 11.5, 3.0 Hz, 2 H), 2.50 (s, 3 H).

^{13}C NMR (100 MHz, CDCl_3): δ = 144.1, 138.1, 137.3, 137.3, 136.6, 136.5, 131.1, 126.7, 126.1, 126.0, 125.8, 121.6, 94.6, 60.1, 60.1, 25.9, 25.9, 15.8.

MS (EI): m/z (%) = 386 (M^+ , 46), 357 (22), 356 (100), 328 (7), 323 (36), 69 (5).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{20}\text{H}_{18}\text{O}_2\text{S}_3$: 386.0469; found: 386.0463.

2-(4,4',5,5'-Tetrahydro-7,7'-spirobi[thieno[2,3-c]pyran]-2-yl)pyrimidine (11j)

Cross coupling of spiro compound **1** (100 mg, 0.38 mmol, 1.0 equiv) was performed according to TP2. The intermediate zinc reagent was prepared using 2.48 M *n*-BuLi in hexane (0.17 mL, 0.42 mmol, 1.1 equiv) and 1 M ZnCl_2 in THF solution (0.46 mL, 0.46 mmol, 1.2 equiv) and cross coupling was performed using heteroaryl bromide **10j** (48 mg, 0.30 mmol, 0.8 equiv) and $\text{Pd}(\text{PPh}_3)_4$ (17 mg, 0.015 mmol, 0.05 equiv) at 50 °C for 4 h. Flash column chromatographic purification (silica gel pre-neutralized with Et_3N , isohexane–EtOAc, 75:25) furnished **11j** (99 mg, 96%) as a pale yellow solid; mp 196.0–197.8 °C.

IR (ATR): 2975, 2925, 2874, 1554, 1414, 1391, 1276, 1154, 1034, 973, 933, 803 cm^{-1} .

^1H NMR (400 MHz, CDCl_3): δ = 8.65 (d, J = 4.9 Hz, 2 H), 7.73 (s, 1 H), 7.26 (d, J = 5.1 Hz, 1 H), 7.07 (t, J = 4.9 Hz, 1 H), 6.82 (d, J = 5.1 Hz, 1 H), 4.31 (dddd, J = 13.4, 12.3, 11.7, 3.0 Hz, 2 H), 4.11 (dd, J = 11.1, 5.8 Hz, 2 H), 2.95–3.15 (m, 2 H), 2.67 (ddd, J = 16.2, 6.5, 3.0 Hz, 2 H).

^{13}C NMR (100 MHz, CDCl_3): δ = 161.4, 157.2, 143.0, 142.4, 137.3, 137.0, 136.6, 127.6, 126.0, 125.8, 118.7, 94.6, 60.2, 60.1, 26.0, 25.8.

MS (EI): m/z (%) = 342 (M^+ , 10), 313 (21), 312 (100), 283 (12), 279 (20), 141 (5).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{17}\text{H}_{14}\text{N}_2\text{O}_2\text{S}_2$: 342.0497; found: 342.0490.

2-(Benzo[*b*]thiophen-3-yl)-4,4',5,5'-tetrahydro-7,7'-spirobi[thieno[2,3-c]pyran] (11k)

Cross coupling of spiro compound **1** (100 mg, 0.38 mmol, 1.0 equiv) was performed according to TP2. The intermediate zinc reagent was prepared using 2.48 M *n*-BuLi in hexane (0.17 mL, 0.42 mmol, 1.1 equiv) and 1 M ZnCl_2 in THF solution (0.46 mL, 0.46 mmol, 1.2 equiv) and cross coupling was performed using heteroaryl bromide **10k** (64 mg, 0.30 mmol, 0.8 equiv) and $\text{Pd}(\text{PPh}_3)_4$ (17 mg, 0.015 mmol, 0.05 equiv) at 50 °C for 4 h. Flash column chromatographic purification (silica gel pre-neutralized with Et_3N , isohexane–EtOAc, 75:25) furnished **11k** (103 mg, 87%) as a pale yellow solid; mp 158.4–160.3 °C.

IR (ATR): 3071, 2966, 2903, 1557, 1458, 1422, 1278, 1243, 1157, 1044, 758, 720 cm^{-1} .

^1H NMR (400 MHz, CDCl_3): δ = 8.03 (d, J = 7.6 Hz, 1 H), 7.79 (d, J = 7.4 Hz, 1 H), 7.26–7.40 (m, 3 H), 7.19 (t, J = 5.0 Hz, 1 H), 6.97 (s, 1 H), 6.75 (d, J = 5.1 Hz, 1 H), 4.26 (dddd, J = 13.5, 11.9, 11.7, 3.0 Hz, 2 H), 4.04 (dd, J = 10.6, 5.9 Hz, 2 H), 2.88–3.08 (m, 2 H), 2.59 (ddd, J = 16.0, 9.8, 3.0 Hz, 2 H).

^{13}C NMR (100 MHz, CDCl_3): δ = 140.6, 137.7, 137.3, 137.1, 136.9, 136.5, 130.4, 126.0, 125.9, 124.7, 124.6, 124.1, 123.9, 123.1, 122.9, 94.6, 60.1, 60.1, 25.9, 25.9.

MS (EI): m/z (%) = 396 (M^+ , 45), 366 (100), 333 (33), 290 (6), 184 (5), 158 (5).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{21}\text{H}_{16}\text{O}_2\text{S}_3$: 396.0312; found: 396.0301.

5-Methyl-2-(4,4',5,5'-tetrahydro-7,7'-spirobi[thieno[2,3-c]pyran]-2-yl)pyridine (11l)

Cross coupling of spiro compound **1** (100 mg, 0.38 mmol, 1.0 equiv) was performed according to TP2. The intermediate zinc reagent was prepared using 2.48 M *n*-BuLi in hexane (0.17 mL, 0.42 mmol, 1.1 equiv) and 1 M ZnCl_2 in THF solution (0.46 mL, 0.46 mmol, 1.2 equiv) and cross coupling was performed using heteroaryl bromide **10l** (52

mg, 0.30 mmol, 0.8 equiv) and $\text{Pd}(\text{PPh}_3)_4$ (17 mg, 0.015 mmol, 0.05 equiv) at 50 °C for 6 h. Flash column chromatographic purification (silica gel pre-neutralized with Et_3N , isohexane– EtOAc , 67:33) furnished **11i** (91 mg, 85%) as a pale yellow solid; mp 202.1–203.1 °C.

IR (ATR): 2964, 2876, 1595, 1498, 1458, 1276, 1161, 1028, 973, 932, 828, 712 cm^{-1} .

^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ = 8.28 (s, 1 H), 7.75 (d, J = 8.0 Hz, 1 H), 7.58–7.66 (m, 1 H), 7.44–7.53 (m, 2 H), 6.90 (d, J = 5.1 Hz, 1 H), 3.96–4.16 (m, 4 H), 2.78–2.93 (m, 2 H), 2.58–2.70 (m, 2 H), 2.27 (s, 3 H).

^{13}C NMR (100 MHz, $\text{DMSO}-d_6$): δ = 149.9, 149.4, 144.7, 139.2, 137.9, 137.5, 137.4, 136.9, 132.4, 126.5, 126.4, 123.6, 118.5, 94.4, 59.8, 59.7, 25.8, 25.7, 18.1.

MS (EI): m/z (%) = 355 (M^+ , 15), 325 (100), 296 (20), 292 (14), 264 (6), 148 (6).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{19}\text{H}_{17}\text{NO}_2\text{S}_2$: 355.0701; found: 355.0697.

2-Methyl-5-(4,4',5,5'-tetrahydro-7,7'-spirobi[thieno[2,3-c]pyran]-2-yl)benzothiazole (11m)

Cross coupling of spiro compound **1** (100 mg, 0.38 mmol, 1.0 equiv) was performed according to TP2. The intermediate zinc reagent was prepared using 2.48 M *n*-BuLi in hexane (0.17 mL, 0.42 mmol, 1.1 equiv) and 1 M ZnCl_2 in THF solution (0.46 mL, 0.46 mmol, 1.2 equiv) and cross coupling was performed using heteroaryl bromide **10m** (69 mg, 0.30 mmol, 0.8 equiv) and $\text{Pd}(\text{PPh}_3)_4$ (17 mg, 0.015 mmol, 0.05 equiv) at 50 °C for 5 h. Flash column chromatographic purification (silica gel pre-neutralized with Et_3N , isohexane– EtOAc , 56:44) furnished **11m** (106 mg, 86%) as a pale yellow solid; mp 243.1–244.0 °C.

IR (ATR): 3096, 2969, 2875, 1518, 1422, 1276, 1159, 1033, 974, 903, 808, 745 cm^{-1} .

^1H NMR (600 MHz, CDCl_3): δ = 8.10 (s, 1 H), 7.73 (d, J = 8.2 Hz, 1 H), 7.50 (d, J = 8.2 Hz, 1 H), 7.25 (d, J = 1.9 Hz, 1 H), 7.07 (s, 1 H), 6.82 (d, J = 4.9 Hz, 1 H), 4.30 (dddd, J = 14.8, 11.8, 11.5, 3.0 Hz, 2 H), 4.09 (dd, J = 11.0, 5.2 Hz, 2 H), 2.96–3.10 (m, 2 H), 2.82 (s, 3 H), 2.57–2.71 (m, 2 H).

^{13}C NMR (150 MHz, CDCl_3): δ = 168.1, 153.6, 144.0, 137.4, 137.2, 137.1, 136.5, 134.7, 132.5, 125.9, 125.8, 123.0, 122.3, 121.6, 119.1, 94.5, 60.1, 60.0, 25.9, 25.8, 20.1.

MS (EI): m/z (%) = 411 (M^+ , 23), 381 (100), 353 (11), 348 (37), 277 (12), 181 (6).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{21}\text{H}_{17}\text{NO}_2\text{S}_3$: 411.0421; found: 411.0411.

(4-tert-Butylphenyl)(4,4',5,5'-tetrahydro-7,7'-spirobi[thieno[2,3-c]pyran]-2-yl)methanone (11n)

Cross coupling of spiro compound **1** (100 mg, 0.38 mmol, 1.0 equiv) was performed according to TP2. The intermediate zinc reagent was prepared using 2.48 M *n*-BuLi in hexane (0.17 mL, 0.42 mmol, 1.1 equiv) and 1 M ZnCl_2 in THF solution (0.46 mL, 0.46 mmol, 1.2 equiv) and cross coupling was performed using the acid chloride **10n** (59 mg, 0.30 mmol, 0.8 equiv) and $\text{Pd}(\text{PPh}_3)_4$ (17 mg, 0.015 mmol, 0.05 equiv) at 50 °C for 3 h. Flash column chromatographic purification (silica gel pre-neutralized with Et_3N , isohexane– EtOAc , 65:35) furnished **11n** (120 mg, 94%) as a white solid; mp 139.1–143.3 °C.

IR (ATR): 3092, 2963, 1628, 1604, 1555, 1456, 1274, 1244, 1162, 1028, 972, 935, 738, 704 cm^{-1} .

^1H NMR (400 MHz, CDCl_3): δ = 7.78 (d, J = 8.4 Hz, 2 H), 7.47 (d, J = 8.4 Hz, 2 H), 7.39 (s, 1 H), 7.24 (d, J = 2.7 Hz, 1 H), 6.81 (d, J = 5.1 Hz, 1 H), 4.27 (dddd, J = 13.4, 12.1, 11.3, 3.2 Hz, 2 H), 4.08 (dd, J = 11.2, 5.7 Hz, 2 H), 2.94–3.07 (m, 2 H), 2.57–2.69 (m, 2 H), 1.34 (s, 9 H).

^{13}C NMR (100 MHz, CDCl_3): δ = 187.9, 156.0, 146.3, 143.0, 136.9, 136.8, 136.4, 135.3, 133.1, 129.2, 126.2, 125.9, 125.4, 94.6, 60.3, 59.9, 35.1, 31.2, 25.9, 25.7.

MS (EI): m/z (%) = 424 (M^+ , 3), 394 (100), 189 (14), 161 (20), 145 (9), 91 (6).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{24}\text{H}_{24}\text{O}_3\text{S}_2$: 424.1167; found: 424.1160.

(4,4',5,5'-Tetrahydro-7,7'-spirobi[thieno[2,3-c]pyran]-2-yl)(thiophen-2-yl)methanone (11o)

Cross coupling of spiro compound **1** (100 mg, 0.38 mmol, 1.0 equiv) was performed according to TP2. The intermediate zinc reagent was prepared using 2.48 M *n*-BuLi in hexane (0.17 mL, 0.42 mmol, 1.1 equiv) and 1 M ZnCl_2 in THF solution (0.46 mL, 0.46 mmol, 1.2 equiv) and cross coupling was performed using the acid chloride **10o** (44 mg, 0.30 mmol, 0.8 equiv) and $\text{Pd}(\text{PPh}_3)_4$ (17 mg, 0.015 mmol, 0.05 equiv) at 50 °C for 4 h. Flash column chromatographic purification (silica gel pre-neutralized with Et_3N , isohexane– EtOAc , 67:33) furnished **11o** (100 mg, 89%) as a pale yellow solid; mp 155.4–157.7 °C.

IR (ATR): 3093, 2931, 2874, 1623, 1552, 1514, 1412, 1274, 1160, 1049, 975, 775, 724 cm^{-1} .

^1H NMR (400 MHz, CDCl_3): δ = 7.89 (d, J = 3.3 Hz, 1 H), 7.69 (d, J = 4.9 Hz, 1 H), 7.64 (s, 1 H), 7.25–7.34 (m, 1 H), 7.17 (t, J = 4.4 Hz, 1 H), 6.85 (d, J = 5.1 Hz, 1 H), 4.32 (dddd, J = 13.1, 12.0, 11.7, 3.0 Hz, 2 H), 4.13 (ddd, J = 11.2, 5.5, 0.8 Hz, 2 H), 2.96–3.17 (m, 2 H), 2.70 (ddd, J = 16.2, 8.4, 3.0 Hz, 2 H).

^{13}C NMR (100 MHz, CDCl_3): δ = 178.8, 146.0, 142.8, 142.0, 137.0, 136.8, 136.4, 133.6, 133.2, 131.9, 128.0, 126.2, 126.0, 94.6, 60.3, 59.9, 25.9, 25.7.

MS (EI): m/z (%) = 374 (M^+ , 4), 344 (100), 234 (5), 110 (43).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{18}\text{H}_{14}\text{O}_3\text{S}_3$: 374.0105; found: 374.0108.

4-[2'-(Methylthio)-4,4',5,5'-tetrahydro-7,7'-spirobi[thieno[2,3-c]pyran]-2-yl]benzonitrile (15a)

Cross coupling of spiro compound **11b** (60 mg, 0.19 mmol, 1.0 equiv) was performed according to TP2. The intermediate zinc reagent was prepared using 2.48 M *n*-BuLi in hexane (0.085 mL, 0.21 mmol, 1.1 equiv) and 1 M ZnCl_2 in THF solution (0.23 mL, 0.23 mmol, 1.2 equiv) and cross coupling was performed using the aryl bromide **10g** (28 mg, 0.15 mmol, 0.8 equiv) and $\text{Pd}(\text{PPh}_3)_4$ (9 mg, 0.008 mmol, 0.05 equiv) at 50 °C for 3 h. Flash column chromatographic purification (silica gel pre-neutralized with Et_3N , isohexane– EtOAc , 67:33) furnished **15a** (46 mg, 75%) as a pale yellow solid; mp 158.4–159.6 °C.

IR (ATR): 2973, 2954, 2881, 2227, 1602, 1558, 1511, 1268, 1153, 1040, 980, 930, 820 cm^{-1} .

^1H NMR (400 MHz, CDCl_3): δ = 7.47–7.57 (m, 4 H), 7.05 (s, 1 H), 6.70 (s, 1 H), 4.12–4.27 (m, 2 H), 4.00 (td, J = 10.6, 5.8 Hz, 2 H), 2.81–3.00 (m, 2 H), 2.53 (ddd, J = 16.2, 7.6, 2.5 Hz, 2 H), 2.37 (s, 3 H).

^{13}C NMR (100 MHz, CDCl_3): δ = 142.2, 139.1, 139.0, 138.4, 137.8, 136.9, 132.7, 128.9, 126.0, 124.0, 118.8, 110.9, 94.1, 60.1, 60.0, 25.8, 25.7, 21.5.

MS (EI): m/z (%) = 411 (M^+ , 26), 382 (25), 381 (100), 366 (10), 348 (14), 334 (30), 291 (6).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{21}\text{H}_{17}\text{NO}_2\text{S}_3$: 411.0421; found: 411.0417.

2-(4-Cyanophenyl)-2'-[4-(methylthio)phenyl]-4,4',5,5'-tetrahydro-7,7'-spiro[thieno[2,3-c]pyran (15b)

Cross coupling of spiro compound **11i** (100 mg, 0.26 mmol, 1.0 equiv) was performed according to TP2. The intermediate zinc reagent was prepared using 2.48 M *n*-BuLi in hexane (0.11 mL, 0.28 mmol, 1.1 equiv) and 1 M ZnCl₂ in THF solution (0.31 mL, 0.31 mmol, 1.2 equiv) and cross coupling was performed using the aryl bromide **10g** (38 mg, 0.208 mmol, 0.8 equiv) and Pd(PPh₃)₄ (12 mg, 0.01 mmol, 0.05 equiv) at 50 °C for 3 h. Flash column chromatographic purification (silica gel pre-neutralized with Et₃N, isohexane–EtOAc, 50:50) furnished **15b** (82 mg, 80%) as a pale yellow solid; mp 225.0–228.0 °C.

IR (ATR): 2955, 2895, 2222, 1602, 1567, 1465, 1313, 1264, 1162, 1032, 970, 926, 829, 814 cm⁻¹.

¹H NMR (400 MHz, CDCl₃): δ = 7.55–7.64 (m, 4 H), 7.44 (d, *J* = 8.2 Hz, 2 H), 7.20 (d, *J* = 8.2 Hz, 2 H), 7.14 (s, 1 H), 7.00 (s, 1 H), 4.25–4.40 (m, 2 H), 4.02–4.19 (m, 2 H), 2.94–3.13 (m, 2 H), 2.59–2.72 (m, 2 H), 2.48 (s, 3 H).

¹³C NMR (100 MHz, CDCl₃): δ = 144.5, 142.2, 139.3, 138.4, 138.4, 137.9, 137.6, 135.9, 132.7, 130.9, 126.7, 126.1, 126.0, 124.0, 121.7, 118.8, 110.9, 94.3, 60.2, 60.0, 25.9, 25.8, 15.7.

MS (EI): *m/z* (%) = 487 (M⁺, 40), 457 (100), 424 (31), 367 (6), 171 (6).

HRMS (EI): *m/z* [M]⁺ calcd for C₂₇H₂₁NO₂S₃: 487.0734; found: 487.0726.

2-(Benzo[*b*]thiophen-3-yl)-2'-[4-(methylthio)phenyl]-4,4',5,5'-tetrahydro-7,7'-spiro[thieno[2,3-c]pyran (15c)

Cross coupling of spiro compound **11i** (100 mg, 0.26 mmol, 1.0 equiv) was performed according to TP2. The intermediate zinc reagent was prepared using 2.48 M *n*-BuLi in hexane (0.11 mL, 0.28 mmol, 1.1 equiv) and 1 M ZnCl₂ in THF solution (0.31 mL, 0.31 mmol, 1.2 equiv) and cross coupling was performed using heteroaryl bromide **10k** (44 mg, 0.208 mmol, 0.8 equiv) and Pd(PPh₃)₄ (12 mg, 0.01 mmol, 0.05 equiv) at 50 °C for 3 h. Flash column chromatographic purification (silica gel pre-neutralized with Et₃N, isohexane–EtOAc, 60:40) furnished **15c** (73 mg, 68%) as a pale yellow solid; mp 190.7–191.2 °C.

IR (ATR): 3067, 2966, 2918, 1566, 1502, 1462, 1277, 1154, 1050, 1032, 968, 902, 811, 758, 729 cm⁻¹.

¹H NMR (400 MHz, CDCl₃): δ = 8.12 (d, *J* = 7.6 Hz, 1 H), 7.87 (d, *J* = 7.4 Hz, 1 H), 7.34–7.50 (m, 5 H), 7.21 (d, *J* = 8.2 Hz, 2 H), 7.07 (s, 1 H), 7.00 (s, 1 H), 4.26–4.44 (m, 2 H), 4.05–4.21 (m, 2 H), 2.95–3.18 (m, 2 H), 2.59–2.78 (m, 2 H), 2.48 (s, 3 H).

¹³C NMR (100 MHz, CDCl₃): δ = 144.3, 140.6, 138.2, 137.9, 137.5, 137.1, 137.0, 136.6, 136.4, 131.1, 130.4, 126.8, 126.2, 124.7, 124.6, 124.2, 124.0, 123.1, 122.9, 121.7, 94.5, 60.2 (2 C), 26.0, 25.9, 15.8.

MS (EI): *m/z* (%) = 518 (M⁺, 41), 488 (100), 455 (39), 259 (8), 184 (8).

HRMS (EI): *m/z* [M]⁺ calcd for C₂₈H₂₂O₂S₄: 518.0503; found: 518.0498.

2-(5'-Methyl-[2,2'-bithiophen]-5-yl)-4,4',5,5'-tetrahydro-7,7'-spiro[thieno[2,3-c]pyran (11p)

Cross coupling of spiro compound **1** (200 mg, 0.76 mmol, 1.0 equiv) was performed according to TP2. The intermediate zinc reagent was prepared using 2.48 M *n*-BuLi in hexane (0.34 mL, 0.83 mmol, 1.1 equiv) and 1 M ZnCl₂ in THF solution (0.91 mL, 0.91 mmol, 1.2 equiv) and cross coupling was performed using heteroaryl bromide **10p** (155 mg, 0.60 mmol, 0.8 equiv) and Pd(PPh₃)₄ (35 mg, 0.03 mmol, 0.05 equiv) at 50 °C for 3 h. Flash column chromatographic purification (silica gel pre-neutralized with Et₃N, isohexane–EtOAc, 70:30) furnished **11p** (170 mg, 64%) as a yellow solid; mp 191.1–192.1 °C.

IR (ATR): 2968, 2921, 2853, 1558, 1484, 1432, 1275, 1223, 1152, 1032, 974, 901, 799, 711 cm⁻¹.

¹H NMR (600 MHz, CDCl₃): δ = 7.27 (d, *J* = 4.9 Hz, 1 H), 6.91–6.98 (m, 3 H), 6.87 (s, 1 H), 6.83 (d, *J* = 5.2 Hz, 1 H), 6.63–6.68 (m, 1 H), 4.29 (dddd, *J* = 14.8, 12.8, 11.5, 3.0 Hz, 2 H), 4.04–4.12 (m, 2 H), 2.95–3.06 (m, 2 H), 2.62 (ddd, *J* = 16.2, 7.1, 3.0 Hz, 2 H), 2.48 (s, 3 H).

¹³C NMR (150 MHz, CDCl₃): δ = 139.4, 137.5, 137.1, 137.0, 136.5, 136.4, 135.3, 134.7, 126.0, 125.9, 125.9, 124.4, 123.6, 123.5, 122.2, 94.5, 60.1, 60.0, 25.9, 25.8, 15.4.

MS (EI): *m/z* (%) = 442 (M⁺, 77), 412 (100), 379 (45), 288 (6), 183 (8).

HRMS (EI): *m/z* [M]⁺ calcd for C₂₂H₁₈O₂S₄: 442.0190; found: 442.0186.

2-{5'-(Adamantan-1-yl)-[2,2'-bithiophen]-5-yl}-4,4',5,5'-tetrahydro-7,7'-spiro[thieno[2,3-c]pyran (11q)

Cross coupling of spiro compound **1** (100 mg, 0.38 mmol, 1.0 equiv) was performed according to TP2. The intermediate zinc reagent was prepared using 2.48 M *n*-BuLi in hexane (0.17 mL, 0.42 mmol, 1.1 equiv) and 1 M ZnCl₂ in THF solution (0.46 mL, 0.46 mmol, 1.2 equiv) and cross coupling was performed using heteroaryl bromide **10q** (114 mg, 0.30 mmol, 0.8 equiv) and Pd(PPh₃)₄ (17 mg, 0.015 mmol, 0.05 equiv) at 50 °C for 3 h. Flash column chromatographic purification (silica gel pre-neutralized with Et₃N, isohexane–EtOAc, 70:30) furnished **11q** (115 mg, 68%) as a yellow solid; mp 109.1–113.7 °C.

IR (ATR): 2899, 2846, 1555, 1443, 1314, 1277, 1154, 1043, 973, 903, 789, 713 cm⁻¹.

¹H NMR (400 MHz, CDCl₃): δ = 7.21–7.29 (m, 1 H), 6.90–7.00 (m, 3 H), 6.76–6.89 (m, 2 H), 6.69 (d, *J* = 3.7 Hz, 1 H), 4.19–4.36 (m, 2 H), 4.06 (ddd, *J* = 11.0, 5.4, 0.7 Hz, 2 H), 2.90–3.07 (m, 2 H), 2.61 (td, *J* = 16.7, 2.6 Hz, 2 H), 2.02–2.12 (m, 3 H), 1.89–2.00 (m, 6 H), 1.67–1.82 (m, 6 H).

¹³C NMR (100 MHz, CDCl₃): δ = 158.0, 137.6, 137.3, 137.1, 136.5, 136.3, 135.2, 133.5, 125.9, 125.9, 124.5, 123.5, 123.2, 122.2, 121.2, 94.5, 60.1, 60.0, 44.8, 36.6, 36.5, 28.9, 25.9, 25.8.

MS (EI): *m/z* (%) = 562 (M⁺, 91), 532 (100), 499 (34), 265 (6), 220 (8).

HRMS (EI): *m/z* [M]⁺ calcd for C₃₁H₃₀O₂S₄: 562.1129; found: 562.1124.

2-([2,2':5',2''-Terthiophen]-5-yl)-4,4',5,5'-tetrahydro-7,7'-spiro[thieno[2,3-c]pyran (11r)

Cross coupling of spiro compound **1** (100 mg, 0.38 mmol, 1.0 equiv) was performed according to TP2. The intermediate zinc reagent was prepared using 2.48 M *n*-BuLi in hexane (0.17 mL, 0.42 mmol, 1.1 equiv) and 1 M ZnCl₂ in THF solution (0.46 mL, 0.46 mmol, 1.2 equiv) and cross coupling was performed using heteroaryl bromide **10r** (98 mg, 0.30 mmol, 0.8 equiv) and Pd(PPh₃)₄ (17 mg, 0.015 mmol, 0.05 equiv) at 50 °C for 3 h. Flash column chromatographic purification (silica gel pre-neutralized with Et₃N, isohexane–EtOAc, 70:30) furnished **11r** (87 mg, 57%) as a yellow solid; mp 221.3–224.9 °C.

IR (ATR): 3060, 2936, 2878, 1442, 1424, 1276, 1223, 1200, 1044, 1034, 975, 938, 910, 831, 821, 788, 703 cm⁻¹.

¹H NMR (400 MHz, CDCl₃): δ = 7.16–7.34 (m, 3 H), 6.97–7.14 (m, 5 H), 6.79–6.96 (m, 2 H), 4.24–4.41 (m, 2 H), 4.02–4.20 (m, 2 H), 2.92–3.14 (m, 2 H), 2.56–2.76 (m, 2 H).

¹³C NMR (100 MHz, CDCl₃): δ = 137.3, 137.2, 137.1, 136.7, 136.6, 136.4, 136.2, 136.1, 135.8, 127.9, 126.0, 125.9, 124.6, 124.4, 124.3, 124.3, 123.8, 122.5, 94.5, 60.1, 60.0, 25.9, 25.8.

MS (EI): *m/z* (%) = 509 (M⁺, 97), 480 (100), 446 (41), 223 (8), 126 (5).

HRMS (EI): *m/z* [M]⁺ calcd for C₂₅H₁₈O₂S₅: 509.9910; found: 509.9904.

2-(5''-Methyl-[2,2':5',2''-terthiophen]-5-yl)-4,4',5,5'-tetrahydro-7,7'-spirobi[thieno[2,3-c]pyran] (11s)

Cross coupling of spiro compound **1** (200 mg, 0.76 mmol, 1.0 equiv) was performed according to TP2. The intermediate zinc reagent was prepared using 2.48 M *n*-BuLi in hexane (0.34 mL, 0.83 mmol, 1.1 equiv) and 1 M ZnCl₂ in THF solution (0.91 mL, 0.91 mmol, 1.2 equiv) and cross coupling was performed using heteroaryl bromide **10s** (205 mg, 0.60 mmol, 0.8 equiv) and Pd(PPh₃)₄ (35 mg, 0.03 mmol, 0.05 equiv) at 50 °C for 3 h. Flash column chromatographic purification (silica gel pre-neutralized with Et₃N, isohexane–EtOAc, 67:33) furnished **11s** (188 mg, 60%) as a yellow solid; mp 208.6–210.9 °C.

IR (ATR): 3057, 2939, 2910, 2881, 1553, 1519, 1441, 1276, 1224, 1151, 1068, 1031, 972, 937, 911, 793, 714 cm⁻¹.

¹H NMR (600 MHz, CDCl₃): δ = 7.28 (d, *J* = 4.9 Hz, 1 H), 6.93–7.04 (m, 5 H), 6.88 (s, 1 H), 6.83 (d, *J* = 5.2 Hz, 1 H), 6.64–6.69 (m, 1 H), 4.29 (dddd, *J* = 14.2, 12.0, 11.2, 2.7 Hz, 2 H), 4.09 (ddd, *J* = 11.0, 6.0, 0.6 Hz, 2 H), 2.94–3.07 (m, 2 H), 2.64 (ddd, *J* = 16.4, 7.9, 2.7 Hz, 2 H), 2.48 (s, 3 H).

¹³C NMR (150 MHz, CDCl₃): δ = 139.5, 137.3, 137.1, 137.0, 136.9, 136.6, 136.5, 136.3, 135.8, 135.1, 134.7, 126.0, 125.9, 125.9, 124.5, 124.2, 124.0, 123.7, 123.6, 122.4, 94.5, 60.1, 60.0, 25.9, 25.8, 15.4.

MS (EI): *m/z* (%) = 524 (M⁺, 100), 495 (21), 493 (80), 461 (29), 247 (12).

HRMS (EI): *m/z* [M]⁺ calcd for C₂₆H₂₀O₂S₅: 524.0067; found: 524.0071.

2,2'-Bis(5'-methyl-[2,2'-bithiophen]-5-yl)-4,4',5,5'-tetrahydro-7,7'-spirobi[thieno[2,3-c]pyran] (16)

Cross coupling of spiro compound **11p** (80 mg, 0.18 mmol, 1.0 equiv) was performed according to TP2. The intermediate zinc reagent was prepared using 2.48 M *n*-BuLi in hexane (0.08 mL, 0.2 mmol, 1.1 equiv) and 1 M ZnCl₂ in THF solution (0.22 mL, 0.22 mmol, 1.2 equiv) and cross coupling was performed using heteroaryl bromide **10p** (36 mg, 0.14 mmol, 0.8 equiv) and Pd(PPh₃)₄ (8.4 mg, 0.007 mmol, 0.05 equiv) at 50 °C for 5 h. Flash column chromatographic purification (silica gel pre-neutralized with Et₃N, isohexane–EtOAc, 60:40) furnished **16** (55 mg, 63%) as a yellow solid; mp 223.0–224.5 °C.

IR (ATR): 3062, 2913, 2852, 1568, 1527, 1444, 1274, 1207, 1151, 1044, 1032, 971, 912, 798, 785 cm⁻¹.

¹H NMR (600 MHz, CDCl₃): δ = 6.97 (d, *J* = 3.6 Hz, 2 H), 6.94 (d, *J* = 3.0 Hz, 4 H), 6.87 (s, 2 H), 6.61–6.68 (m, 2 H), 4.29 (td, *J* = 11.3, 2.5 Hz, 2 H), 4.09 (dd, *J* = 11.0, 5.7 Hz, 2 H), 2.95–3.04 (m, 2 H), 2.61 (dd, *J* = 16.0, 2.5 Hz, 2 H), 2.47 (s, 6 H).

¹³C NMR (150 MHz, CDCl₃): δ = 139.4, 137.7, 137.2, 137.0, 135.9, 135.2, 134.7, 126.0, 124.5, 123.6, 123.5, 122.1, 94.2, 60.0, 25.8, 15.4.

MS (EI): *m/z* (%) = 620 (M⁺, 90), 590 (100), 560 (26), 557 (69), 272 (8).

HRMS (EI): *m/z* [M]⁺ calcd for C₃₁H₂₄O₂S₆: 620.0101; found: 620.0091.

2-(Pyren-1-yl)-4,4',5,5'-tetrahydro-7,7'-spirobi[thieno[2,3-c]pyran] (18)

Cross coupling of spiro compound **1** (100 mg, 0.38 mmol, 1.0 equiv) was performed according to TP2. The intermediate zinc reagent was prepared using 2.48 M *n*-BuLi in hexane (0.17 mL, 0.42 mmol, 1.1 equiv) and 1 M ZnCl₂ in THF solution (0.46 mL, 0.46 mmol, 1.2 equiv) and cross coupling was performed using the aryl bromide **17** (84 mg, 0.30 mmol, 0.8 equiv) and Pd(PPh₃)₄ (17 mg, 0.015 mmol, 0.05 equiv)

at 50 °C for 3 h. Flash column chromatographic purification (silica gel pre-neutralized with Et₃N, isohexane–EtOAc, 60:40) furnished **18** (101 mg, 73%) as a yellow solid; mp 216.7–217.4 °C.

IR (ATR): 3038, 2974, 2940, 1600, 1560, 1428, 1274, 1248, 1031, 977, 904, 842, 760, 723, 701 cm⁻¹.

¹H NMR (600 MHz, CDCl₃): δ = 8.54 (d, *J* = 9.3 Hz, 1 H), 8.15–8.22 (m, 2 H), 8.13 (d, *J* = 8.0 Hz, 1 H), 7.96–8.10 (m, 5 H), 7.31 (d, *J* = 5.1 Hz, 1 H), 7.09 (s, 1 H), 6.85 (d, *J* = 5.1 Hz, 1 H), 4.46 (td, *J* = 11.6, 2.7 Hz, 1 H), 4.36 (td, *J* = 11.6, 2.7 Hz, 1 H), 4.12–4.25 (m, 2 H), 3.17 (ddd, *J* = 15.8, 12.4, 5.9 Hz, 1 H), 3.07 (ddd, *J* = 15.8, 12.4, 5.9 Hz, 1 H), 2.76 (dd, *J* = 16.0, 2.7 Hz, 1 H), 2.69 (dd, *J* = 16.0, 2.7 Hz, 1 H).

¹³C NMR (150 MHz, CDCl₃): δ = 142.9, 138.3, 137.4, 136.7, 136.5, 131.4, 131.0, 130.9, 129.5, 128.9, 128.2, 127.9, 127.8, 127.3, 126.8, 126.1, 125.9, 125.8, 125.3, 125.0, 125.0, 124.9, 124.7, 124.5, 94.7, 60.1 (2 C), 26.0, 25.9.

MS (EI): *m/z* (%) = 464 (M⁺, 67), 434 (100), 401 (28), 294 (10), 226 (21), 43 (11).

HRMS (EI): *m/z* [M]⁺ calcd for C₂₉H₂₀O₂S₂: 464.0905; found: 464.0905.

2,2'-Di(pyren-1-yl)-4,4',5,5'-tetrahydro-7,7'-spirobi[thieno[2,3-c]pyran] (19)

Cross coupling of spiro compound **18** (135 mg, 0.29 mmol, 1.0 equiv) was performed according to TP2. The intermediate zinc reagent was prepared using 2.48 M *n*-BuLi in hexane (0.13 mL, 0.32 mmol, 1.1 equiv) and 1 M ZnCl₂ in THF solution (0.35 mL, 0.35 mmol, 1.2 equiv) and cross coupling was performed using the aryl bromide **17** (65 mg, 0.23 mmol, 0.8 equiv) and Pd(PPh₃)₄ (13 mg, 0.011 mmol, 0.05 equiv) at 50 °C for 6 h. Flash column chromatographic purification (silica gel pre-neutralized with Et₃N, isohexane–EtOAc, 67:33) furnished **19** (70 mg, 46%) as a yellow solid; mp 257.1–258.5 °C.

IR (ATR): 3041, 2926, 1601, 1566, 1456, 1428, 1271, 1150, 1027, 968, 930, 845, 838, 720, 695 cm⁻¹.

¹H NMR (200 MHz, DMSO-*d*₆): δ = 7.82–8.65 (m, 18 H), 7.30 (s, 2 H), 3.95–4.48 (m, 4 H), 2.65–3.21 (m, 4 H).

¹³C NMR (100 MHz, CDCl₃): δ = 143.0, 138.4, 136.7, 131.4, 131.1, 130.9, 129.6, 129.0, 128.2, 127.9, 127.7, 127.2, 126.7, 126.0, 125.2, 125.0, 125.0, 124.8, 124.4, 94.7, 60.2, 26.0.

MS (EI): *m/z* (%) = 664 (M⁺, 76), 634 (100), 590 (44), 295 (43), 226 (29).

HRMS (EI): *m/z* [M]⁺ calcd for C₄₅H₂₈O₂S₂: 664.1531; found: 664.1538.

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Supporting Information

Supporting information for this article is available online at <http://dx.doi.org/10.1055/s-0035-1560187>.

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